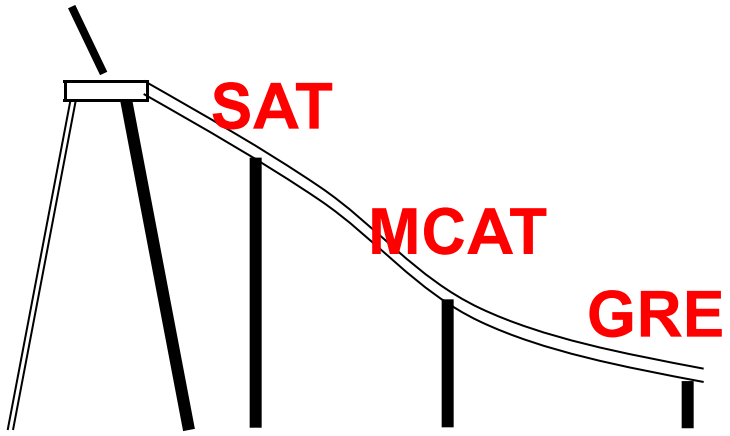
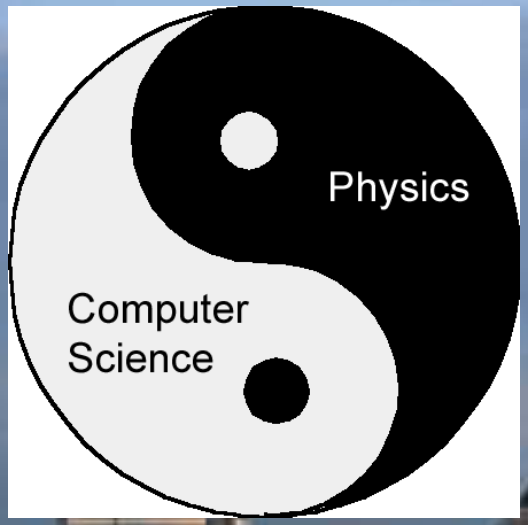
Test Slide



The Power of Quantum Algorithms

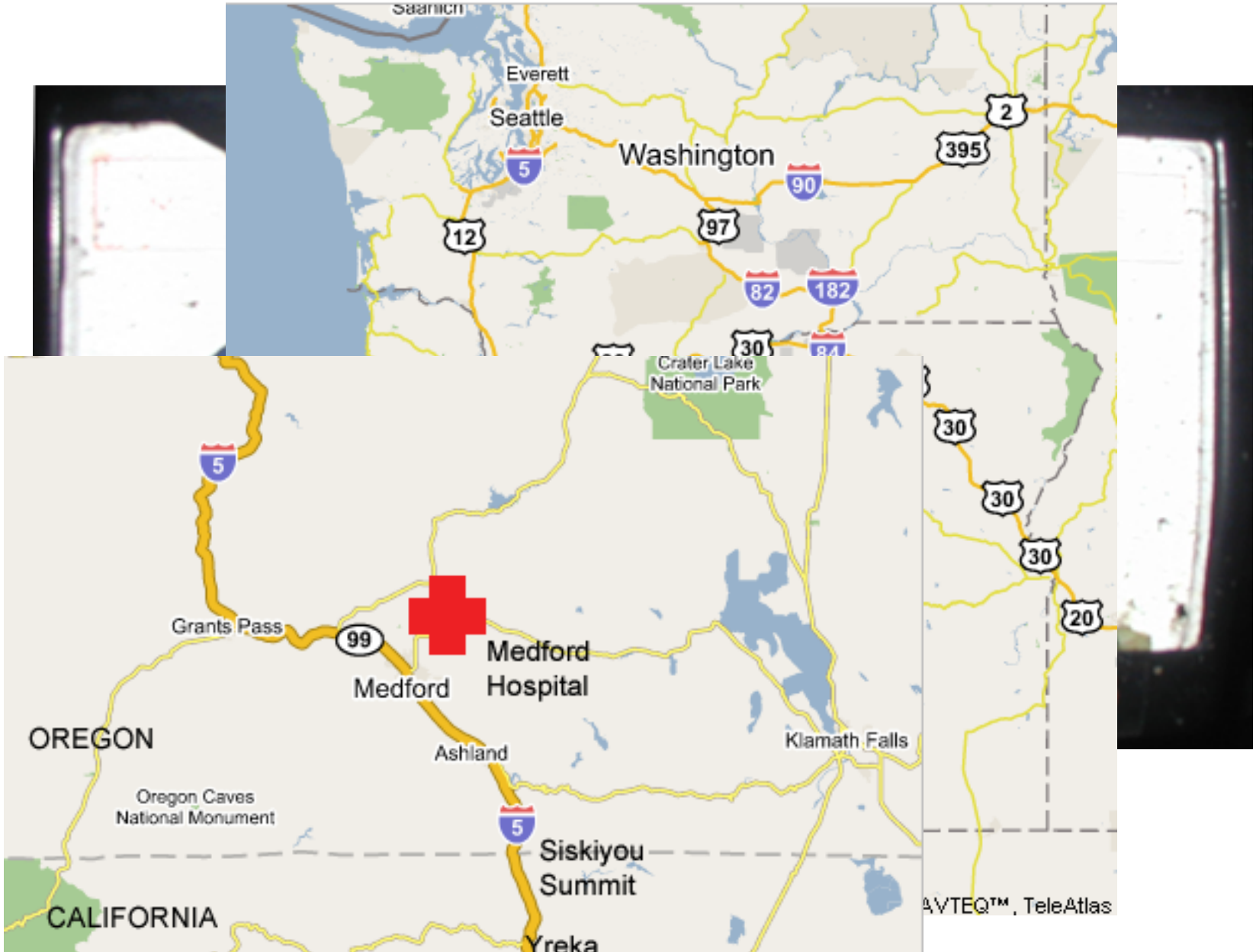




Dave
Ba

Department of Computer Science & Engineering
University of Washington

Apologies


Everett
Seattle
Washington
5
90
2
395
12
97
82
182
30
Crater Lake
National Park
5
Grants Pass
99
Medford
Medford
Hospital
OREGON
Ashland
Klamath Falls
Oregon Caves
National Monument
5
Siskiyou
Summit
CALIFORNIA
Yreka
30
30
30
20
AVTEQ™, TeleAtlas

What Do You See?

SOFTWARE

Email

The Web

Databases

Etc.etc.etc.



HARDWARE

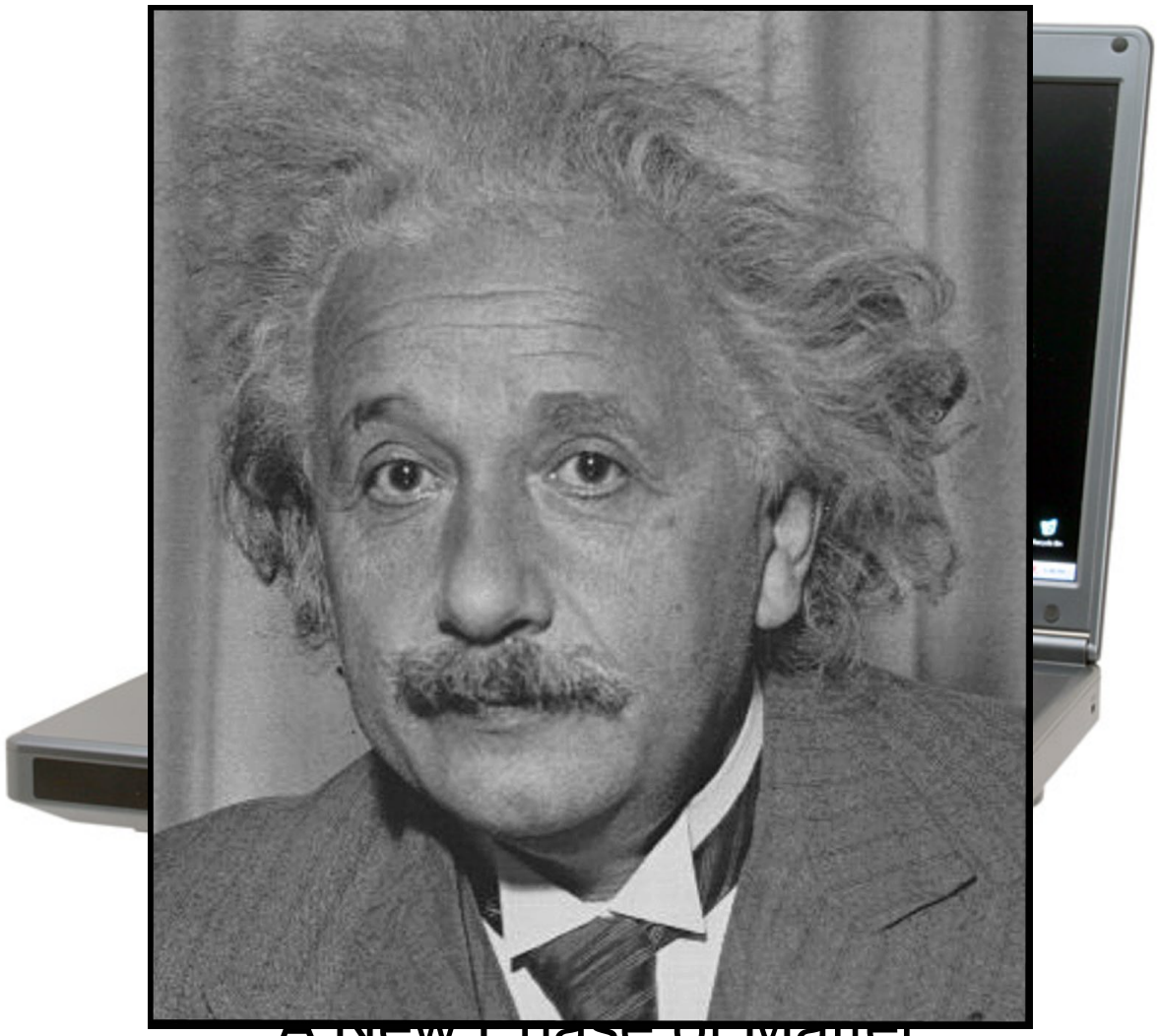
Moore's law

Architecture

Communication

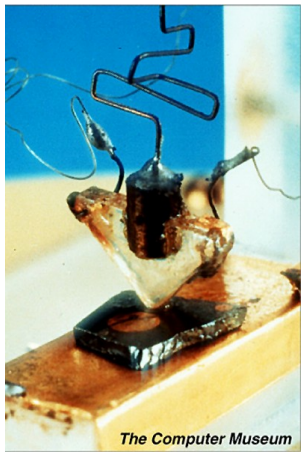
Etc.etc.etc.

What Do Physicists See?



A New Phase of Matter

Computing State of Matter



Transistor



Integrated Circuit

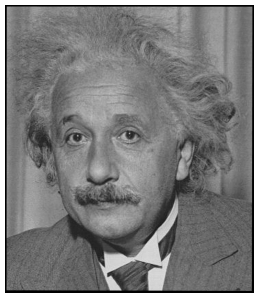


Magnetic Storage Media

Robust (Fault-Tolerant) Storage and Manipulation of Discrete Data

LOOKS VERY LITTLE LIKE

Our fundamental laws of Physics



$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

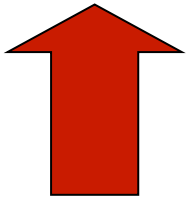
$$i\hbar\frac{\partial\psi}{\partial t} = H\psi$$



Continuous (noisy) data, chaotic, etc.

Maske
d

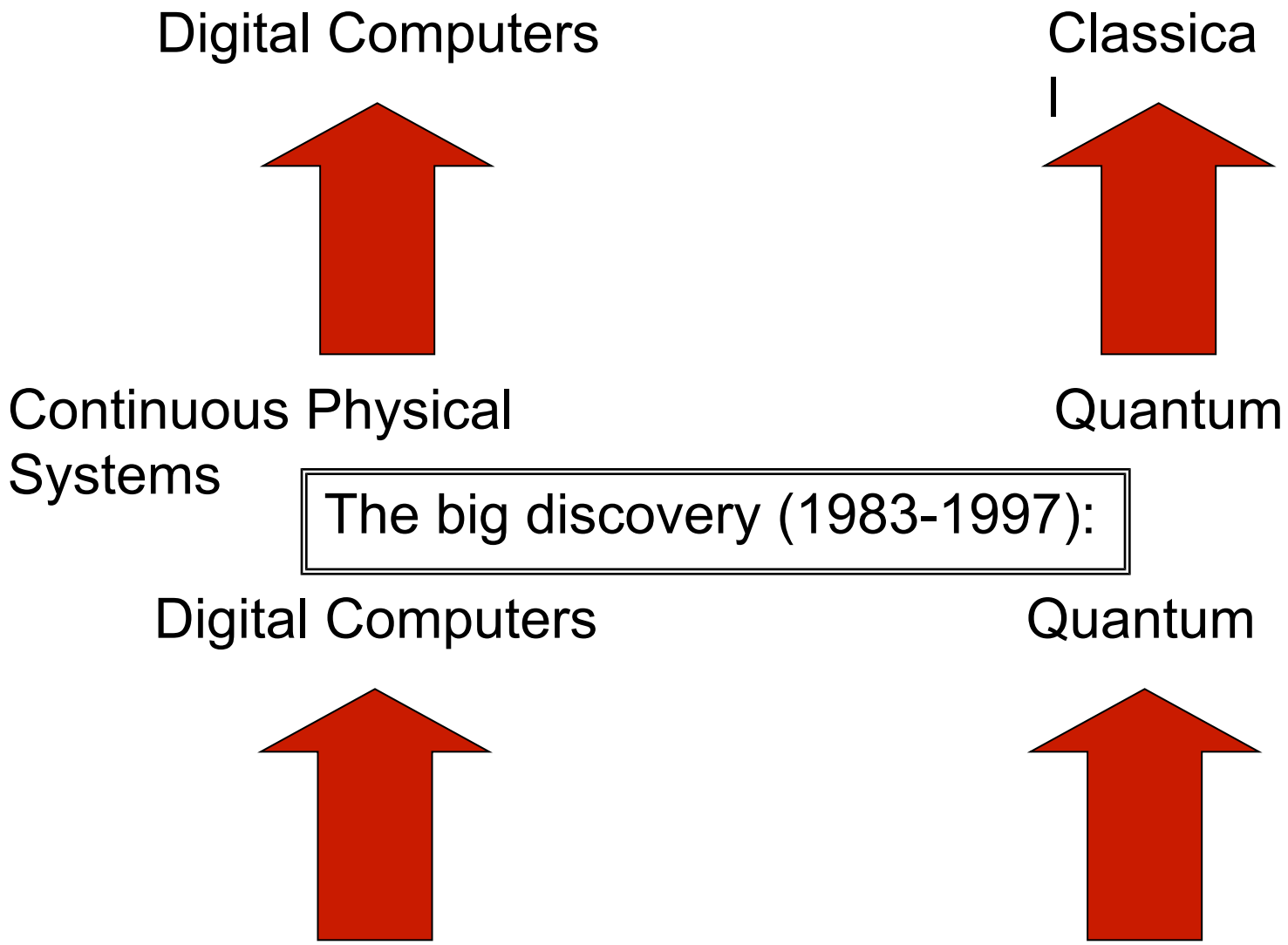
Digital Computers



Noisy, Continuous Physical Systems

What constitutes a computer must ultimately be answered the laws of physics combined with an understanding of how robust behavior can arise in light of this physics.

Two At Once


Digital Computers
Classica
l
Continuous Physical
Systems
Quantum
The big discovery (1983-1997):
Digital Computers
Quantum
Continuous Physical
Quantum

Quantum Computers

Machines whose operation is inherently quantum



Richard Feynman



David Deutsch



Paul Benioff

"How can we simulate quantum mechanics?...Can we do it with a new kind of computer…a quantum computer? It's not a Turing machine but a machine of a different kind" Feynman, 1981

The Factoring Firestorm



Peter Shor 1994

188198812920607963383697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199623212057340318795506569622130516875930765025705 9

=

398075086424064937397125500550386491199064362342526708406385189575946388957261768583317

×

472772146107435302536223071973048224632914695302097116459852171130520711256363590397527

Best classical algorithm takes time $O(\exp(n^{1/3}))$

Shor's quantum algorithm takes time $O(n^2 \log n)$

An efficient algorithm for factoring **breaks** the RSA public key cryptosystem

Reasons to Study

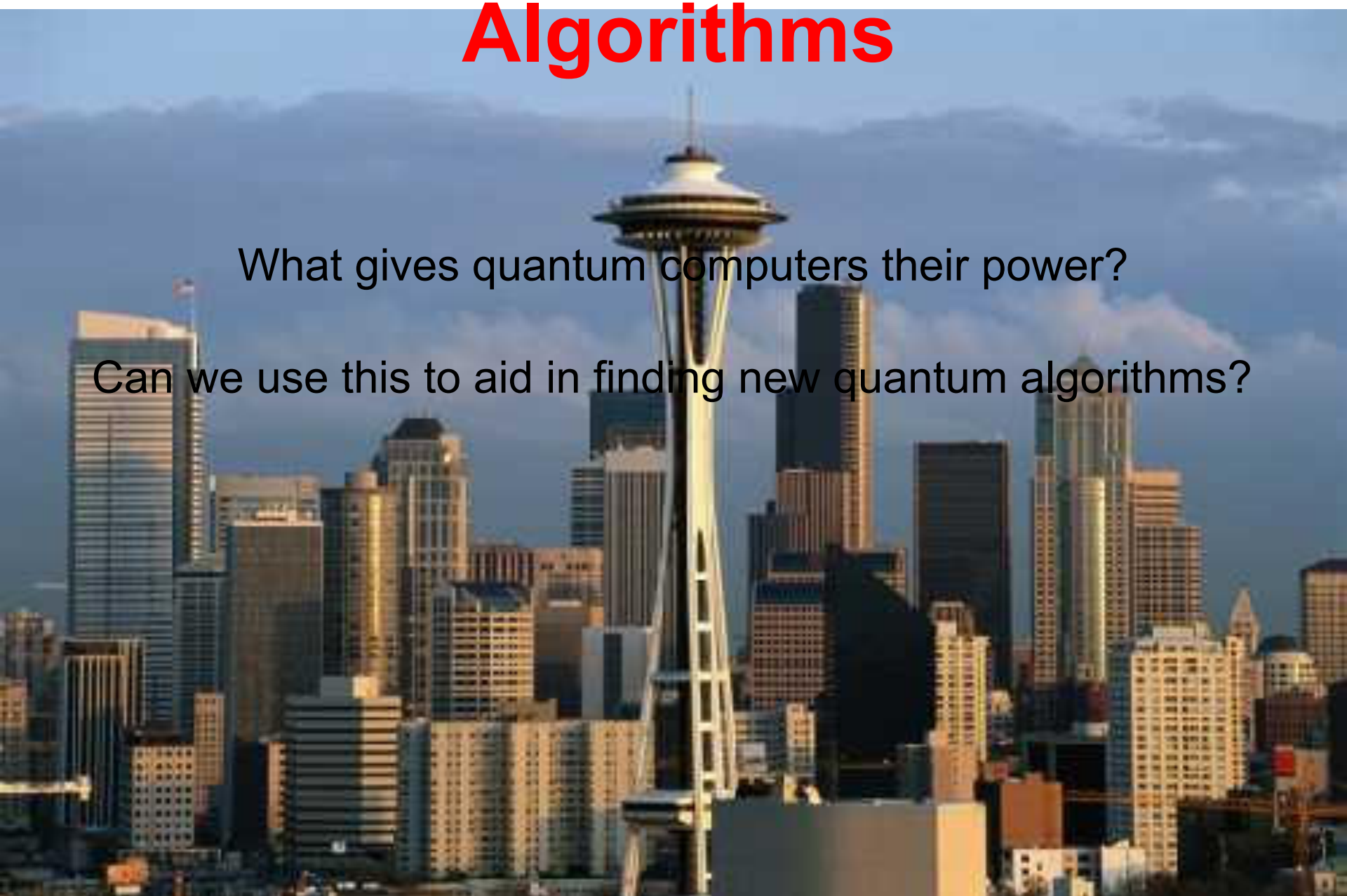
1. In the future we may build large quantum computers.

2. The ultimate limits of computation are quantum limits

   If someone showed P does not equal NP, would we be satisfied today?

3. The beginning of an interplay between physics and computer science?

The Power of Quantum Algorithms

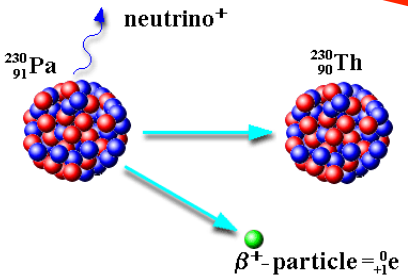


What gives quantum computers their power?

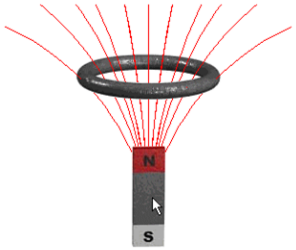
Can we use this to aid in finding new quantum algorithms?

What is Quantum Theory?

Weak Force

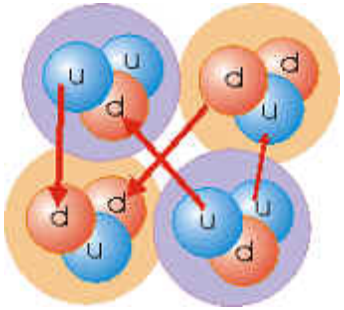


Electromagnetic Force

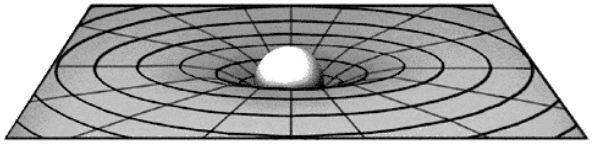




Strong Force



Gravitational Force (?!?!)



Quantum Theory

Quantum theory is the **machine language** of the universe

Bit? Pbit? No: Qubit!

Classical Bit

0

1

two states

Probabilistic Classical Bit

0 *p*

1 *1-p*

two states

description of the system

Quantum Bit

0

1

a 0

a 1

$a_i \in \mathbb{C}$

complex numbers

$|a_0|^2 + |a_1|^2 = 1$

description of the system

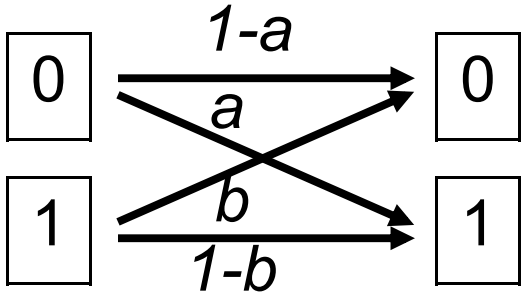
upon measurement, acts like a pbit:

0 *p=|a0|2*

1 *1-p =|a1|2*

Evolution

Probabilistic Classical Bit Evolution



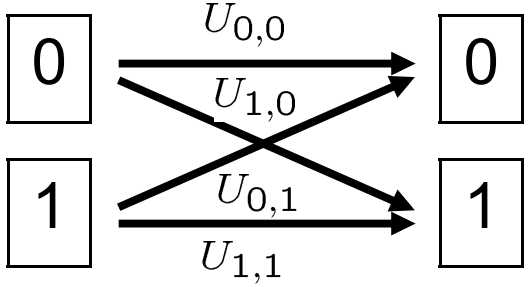
$$\begin{bmatrix} p' \\ 1-p' \end{bmatrix} = \begin{bmatrix} 1-a & b \\ a & 1-b \end{bmatrix} \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

$$\begin{bmatrix} p \\ 1-p \end{bmatrix} \qquad \begin{bmatrix} p' \\ 1-p' \end{bmatrix}$$

$$0 \leq a, b \leq 1$$
$$a, b \in \mathbb{R}$$

stochastic

Quantum Bit Evolution



$$\begin{bmatrix} a'_0 \\ a'_1 \end{bmatrix} = \begin{bmatrix} U_{0,0} & U_{0,1} \\ U_{1,0} & U_{1,1} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$$

$$\begin{bmatrix} a_0 \\ a_1 \end{bmatrix} \qquad \begin{bmatrix} a'_0 \\ a'_1 \end{bmatrix}$$

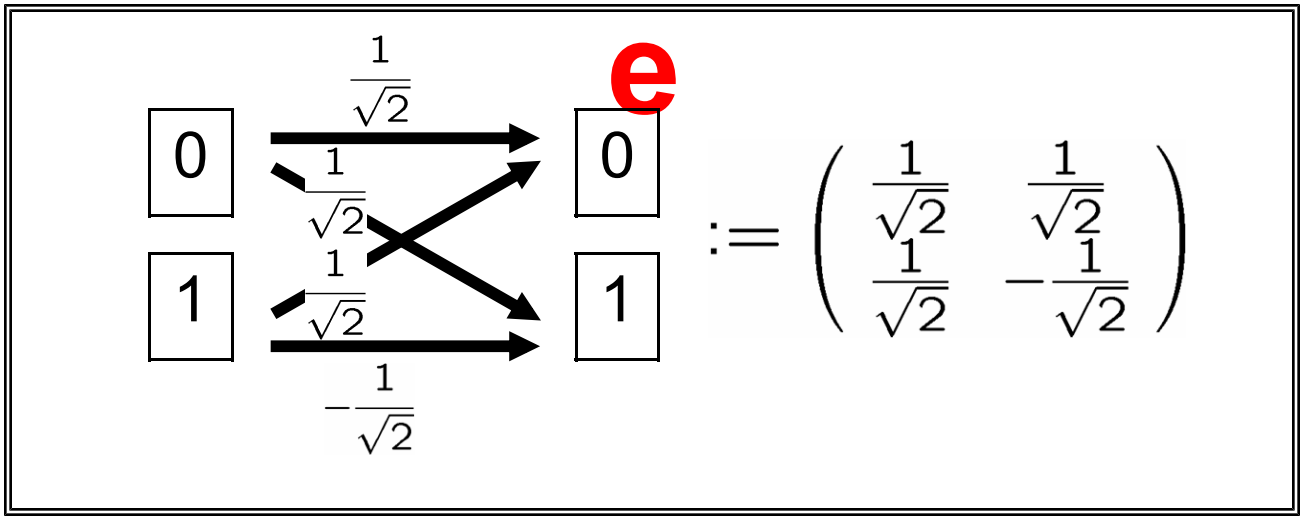
$$U^\dagger = U^{-1}$$
$$U_{i,j} \in \mathbb{C}$$

unitary



Interferenc e





qubit input

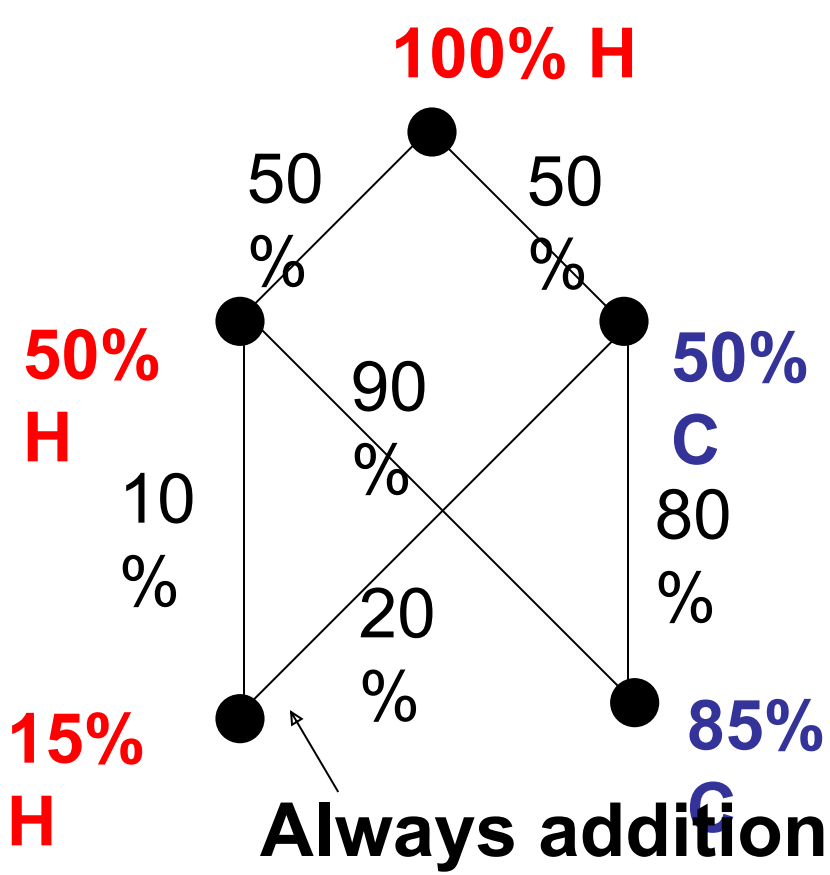
$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ — $\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$ — measure

0	50 %	100%
1	50 %	0%

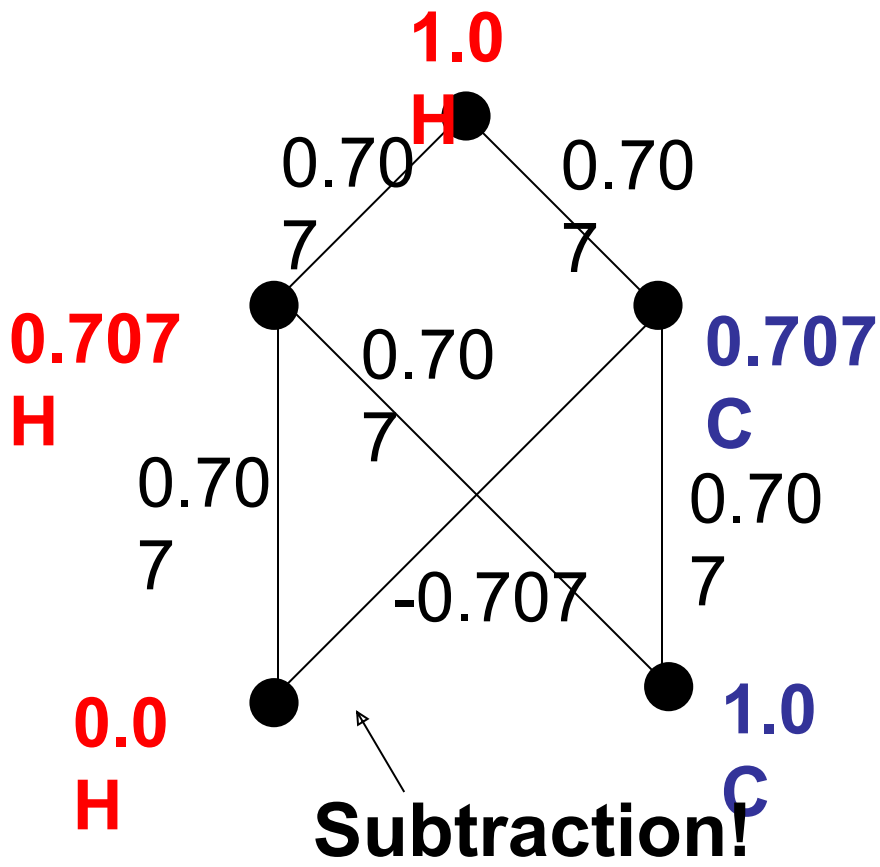
$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$ — $\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$ — measure

0	50 %	0%
1	50 %	100%

Interfering Pathways



Classical



Quantum

Quantum Theory Made Easy

Classical				Quantum			
0	0	0	p0	a0			
0	0	1	p1	a1			
0	1	0	p2	a2			
0	1	1	p3	a3			
1	0	0	p4	a4			
1	0	1	p5	a5			
1	1	0	p6	a6			
1	1	1	p7	a7			

<u>probabilitie</u>

$p_i \in \mathbb{R}$

$0 \leq p_i \leq 1$

$\sum_i p_i = 1$

probability of observing state *s* is *ps*

<u>amplitude</u>

$a_i \in \mathbb{C}$

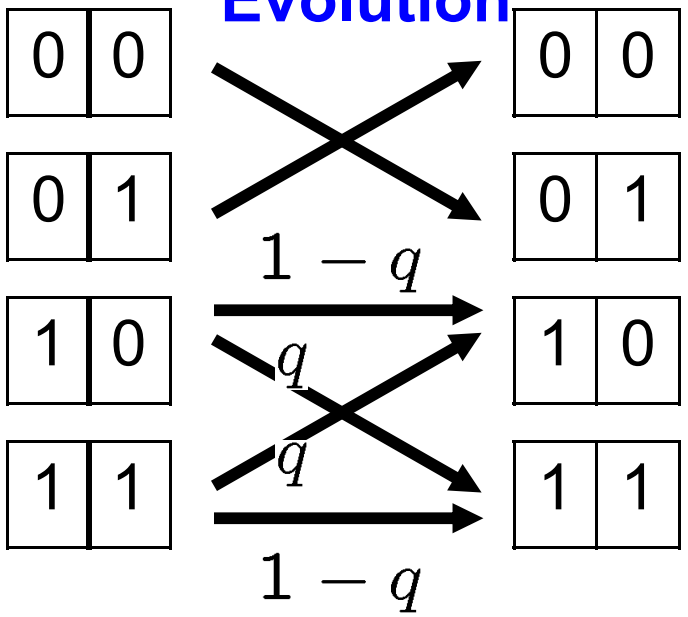
$0 \leq |a_i|^2 \leq 1$

$\sum_i |a_i|^2 = 1$

probability of observing state *s* is |*as*|2

Quantum Theory Made Easy

Classical Evolution

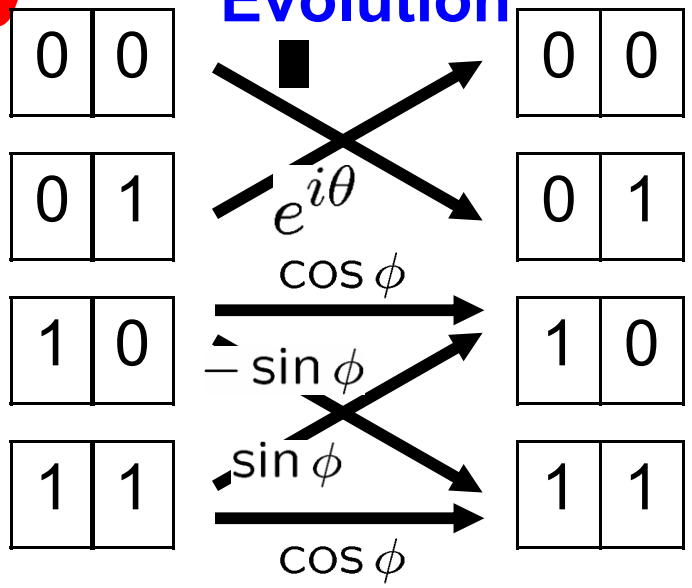


transition probabilities

$$\begin{bmatrix} p'_0 \\ p'_1 \\ p'_2 \\ p'_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1-p & q \\ 0 & 0 & p & 1-q \end{bmatrix} \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix}$$

stochastic matrix

Quantum Evolution



transition amplitudes

$$\begin{bmatrix} a'_0 \\ a'_1 \\ a'_2 \\ a'_3 \end{bmatrix} = \begin{bmatrix} 0 & e^{i\theta} & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & \cos\phi & \sin\phi \\ 0 & 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

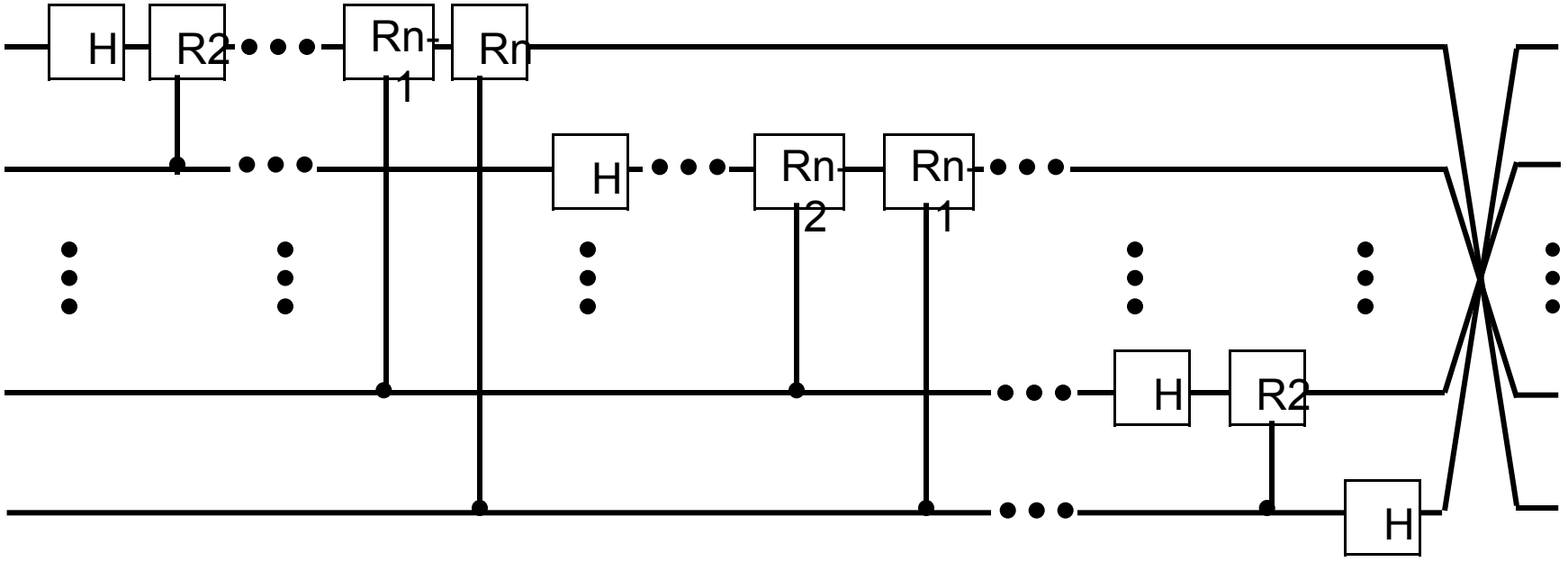
unitary matrix

Quantum Circuits

uniform quantum circuit families

sequence of

- one and two qubit unitary evolutions
- measurements



Quantum circuit for the Fourier Transform over $\mathbb{Z}_{2^n}$

Quantum Algorithms



Umesh Vazirani (U.C. Berkeley)

Quantum computers offer superpolynomial speedups (for certain oracle promise problems) over classical computers [Bernstein and Vazirani STOC 93]

Quantum computers offer exponential speedups (for certain oracle promise problems) over classical computers [Simon FOCS 94]



Dan Simon (Microsoft)



Peter Shor (AT&T→MIT)

Quantum computers can efficiently factor integers and can efficiently compute the discrete logarithm [Shor FOCS 94]

Quantum Algorithms

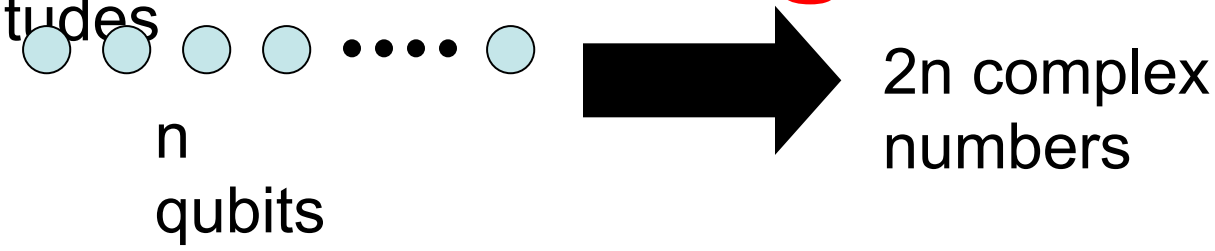
- *Factoring, Discrete log [Shor 94]
- Unstructured search [Grover 96]
- *Abelian hidden subgroup problems [Shor 94, Kitaev 95]
- *Solving Pell's equation [Hallgren 02]
- *Some hidden shift problems [van Dam, Hallgren, Ip 03]
- *Graph traversal [CCDFGS 03]
- Spatial search [AA 03, CG 03/04, AKR 04]
- Element distinctness [Ambainis 03]
- Various graph problems [DHHM 04, MSS 03,…]
- Testing matrix multiplication [Buhrman,Špalek 04]
- …[Your Algorithm Here]

Where is the power of quantum algorithms coming from?

* = suspected exponential gain over classical

Some Counting

To describe the generic state of n qubits, we need 2n amplitudes



But, to describe the generic state of n pbits, we need 2n probabilities



Moral: that our description is exponentially large is cannot be at the heart of the speedup provided by quantum computers. No “massive parallelism for free.”

Separablity



Alice



Bob



Charlie

Generic state requires 8 complex numbers:

$$\begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \end{bmatrix} \begin{matrix} 000 \\ 001 \\ 010 \\ 011 \\ 100 \\ 101 \\ 110 \\ 111 \end{matrix}$$

Separablity

But some states require us to store less numbers:







Alice **Bob** **Charlie**

$$\begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix} \begin{matrix} 0 \\ 1 \end{matrix} \qquad \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \begin{matrix} 0 \\ 1 \end{matrix} \qquad \begin{bmatrix} \gamma_0 \\ \gamma_1 \end{bmatrix} \begin{matrix} 0 \\ 1 \end{matrix}$$

$$\begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix} \otimes \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \otimes \begin{bmatrix} \gamma_0 \\ \gamma_1 \end{bmatrix} = \begin{bmatrix} \alpha_0\beta_0\gamma_0 \\ \alpha_0\beta_0\gamma_1 \\ \alpha_0\beta_1\gamma_0 \\ \alpha_0\beta_1\gamma_1 \\ \alpha_1\beta_0\gamma_0 \\ \alpha_1\beta_0\gamma_1 \\ \alpha_1\beta_1\gamma_0 \\ \alpha_1\beta_1\gamma_1 \end{bmatrix}$$

Entangled States

Some states cannot be expressed as separable states:



Alice

$$\begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix} \begin{matrix} 0 \\ 1 \end{matrix}$$



Bob

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \begin{matrix} 0 \\ 1 \end{matrix}$$



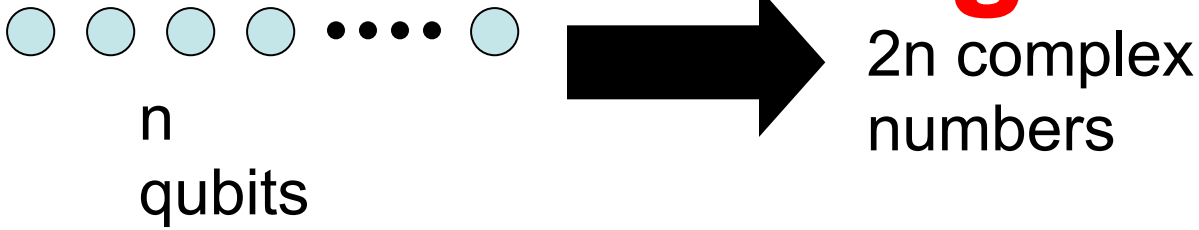
Charlie

$$\begin{bmatrix} \gamma_0 \\ \gamma_1 \end{bmatrix} \begin{matrix} 0 \\ 1 \end{matrix}$$

$$\begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix} \otimes \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \otimes \begin{bmatrix} \gamma_0 \\ \gamma_1 \end{bmatrix} \neq \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

More Counting

n qubits in separable state:



n
qubits

2n complex numbers

Entanglement and **Quantum Speedups**:

small quantity of entanglement → efficient classical algorithm for simulating the system

[Jozsa and Linden, Proc. Math. Phys. & Eng. Sci. 459 2011 (2003)]
[Vidal, Phys. Rev. Lett. 93, 040502 (2004)]

Entanglement is necessary for quantum speedups.

What makes entanglement special?

A Game



Alice

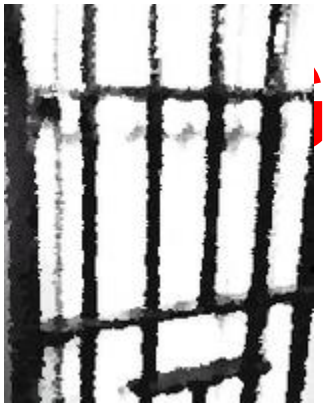


Bob



Charlie

An Evil Game







Alice **Bob** **Charlie**

1. Each given slip **X** or **Y**

2. At midnight each party shouts "+1" or "-1"

Winning the Game







Alice	Bobb	Charlie	Winning Product
X	X	X	+
X	Y	Y	1
Y	Y	X	- 1
Y	X	Y	- 1

A Game







Alice

A

X

A

Y

Bob

B

X

B

Y

Charlie

C

X

C

Y

All either +1 or -1

A Game







Alice	Bob	Charlie	Product
X	X	X	AXBXCX
X	Y	Y	AXBYCY
Y	Y	X	AYBYCX
Y	X	Y	AYBXCY

Winners?

Assuming winning strategy:

AXBXCX = +1

AXBYCY = +1

AYBYCX = +1

AYBXCY = +1

Multiply equations….contradiction!

LHS:

AX2AY2Bx2BY2Cx2CY2=+1

Losers



Alice



Bob



Charlie

There is no strategy to win the game

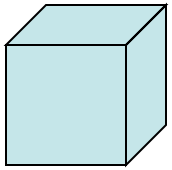
In the Real World



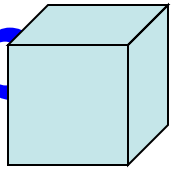




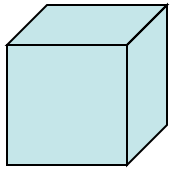
Alice



Bob



Charlie



This is wrong, there is a way to win

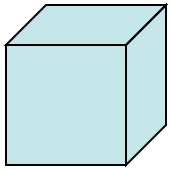
GHZ Sta



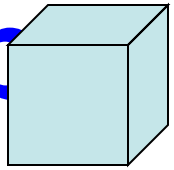




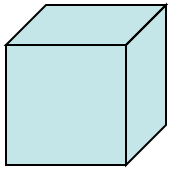
Alic e



Bo b



Charli e



$$\begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix} \otimes \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \otimes \begin{bmatrix} \gamma_0 \\ \gamma_1 \end{bmatrix} \neq \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

GHZ Prot







Alic e

Bo b

Charli e

Each party applies:

$$M_X := \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$M_Y := \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

GHZ Magic







Alic **Bo** **Charli**

XX $\frac{1}{\sqrt{2}}|000\rangle+\frac{1}{\sqrt{2}}|111\rangle \rightarrow \frac{1}{2}|000\rangle+\frac{1}{2}|011\rangle+\frac{1}{2}|101\rangle+\frac{1}{2}|110\rangle$

XX $\frac{1}{\sqrt{2}}|000\rangle+\frac{1}{\sqrt{2}}|111\rangle \rightarrow -\frac{1}{2}|111\rangle+\frac{i}{2}|001\rangle+\frac{1}{2}|100\rangle+\frac{i}{2}|010\rangle$

XY $\frac{1}{\sqrt{2}}|000\rangle+\frac{1}{\sqrt{2}}|111\rangle \rightarrow -\frac{1}{2}|111\rangle+\frac{i}{2}|001\rangle+\frac{i}{2}|100\rangle+\frac{1}{2}|010\rangle$

XX $\frac{1}{\sqrt{2}}|000\rangle+\frac{1}{\sqrt{2}}|111\rangle \rightarrow -\frac{1}{2}|111\rangle+\frac{1}{2}|001\rangle+\frac{i}{2}|100\rangle+\frac{i}{2}|010\rangle$

Y $0 \rightarrow +1 \quad 1 \rightarrow -1$

In the Real World







Alice

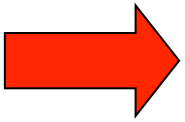
Bob

Charlie

By sharing “entangled” quantum particles, Alice, Bob, and Charlie win!

More Counting

Entanglement and **Quantum Speedups**:

small quantity of entanglement  efficient classical algorithm for simulating the system

[Jozsa and Linden, Proc. Math. Phys. & Eng. Sci. 459 2011 (2003)]
[Vidal, Phys. Rev. Lett. 93, 040502 (2004)]

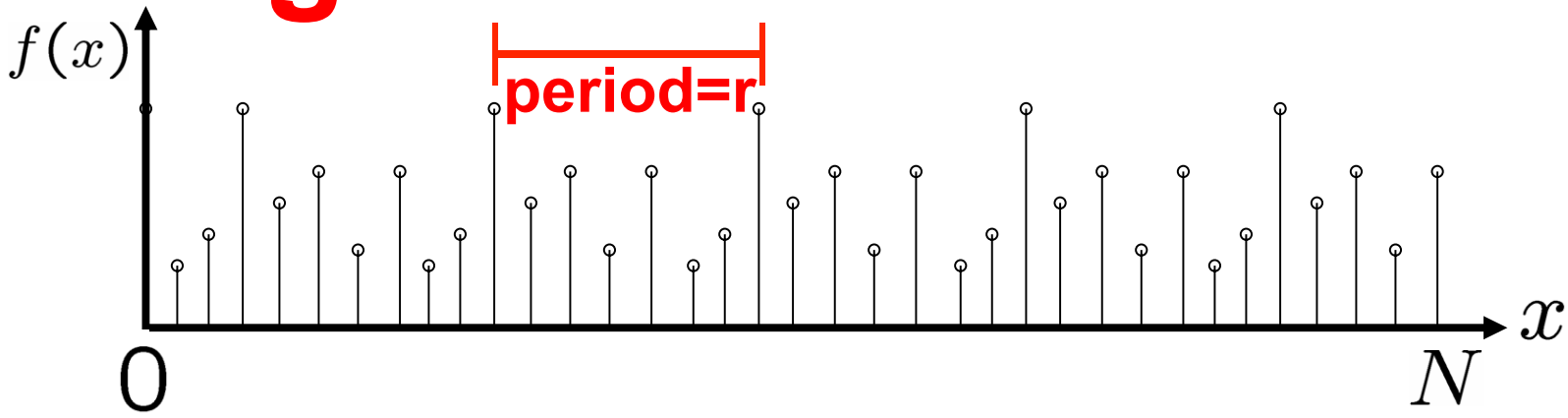
Entanglement is necessary for quantum speedups.

Entanglement is a powerful form of correlation.

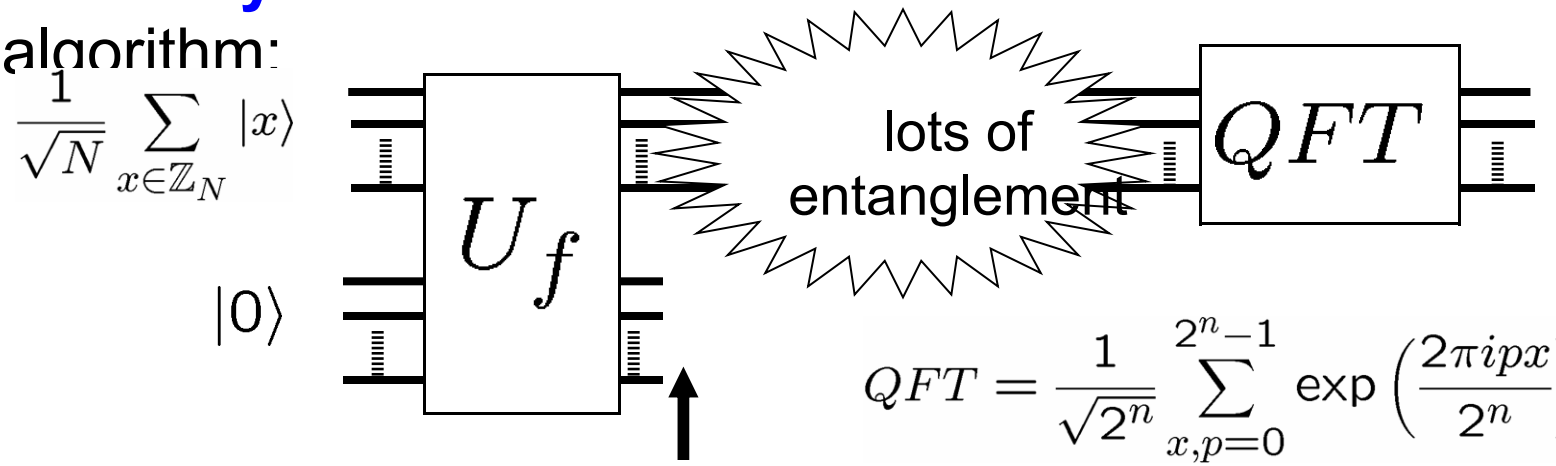
Can we use this to build new quantum algorithms?

Factoring and Entanglement

Factoring can be reduced to period finding.



Anatomy of Shor's algorithm:

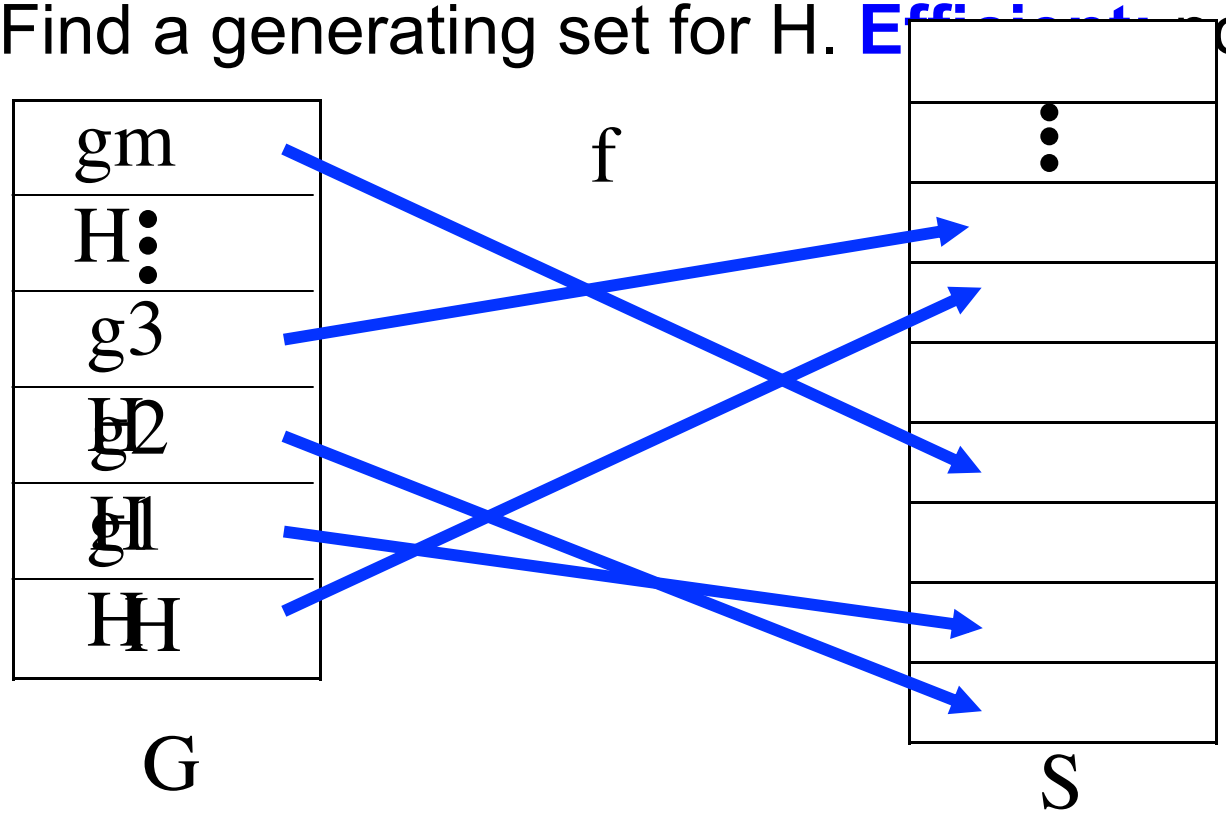


$$QFT = \frac{1}{\sqrt{2^n}} \sum_{x,p=0}^{2^n-1} \exp\left(\frac{2\pi i p x}{2^n}\right) |p\rangle\langle x|$$

Hidden Subgroup Problem

Setup: Fix a group G (known) and a subgroup $H \subseteq G$ (unknown). Given query access to a function $f : G \to S$ that
is constant and distinct on left cosets of H in G. We say that f hides H.

Problem: Find a generating set for H. **Efficient:** polylog(|G|)



Groups

Group Definition

Set of elements with a binary operation "$\cdot$"

Binary operation satisfies:

1. Closure: $g1 \cdot g2$ is in the set
2. Associativity: $(g1 \cdot g2) \cdot g3 = g1 \cdot (g2 \cdot g3)$
3. Identity element: Exists e such that $g1 \cdot e = g1$
4. Inverse: For all g1 there exists an element g2 such that $g1g2 = e$.

We will study only finite groups (set is finite.)

Left Cosets

Subgroups

Subset of group which is itself a group.

Left Cosets

Let H be a subgroup of G.

Def: g1H is the set {g1h, h∈H}

g1H is a left coset of G with respect to H.

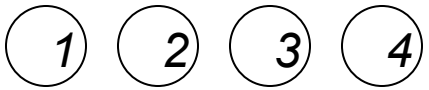
Given group G, subgroup H, we can uniquely partition G into left cosets of H.

Hidden Subgroup Problems

Shor's factoring algorithm is an example of a HSP.

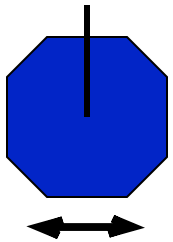
Efficient algorithms for the HSP over

the Symmetric group → an efficient algorithm for graph isomorphism



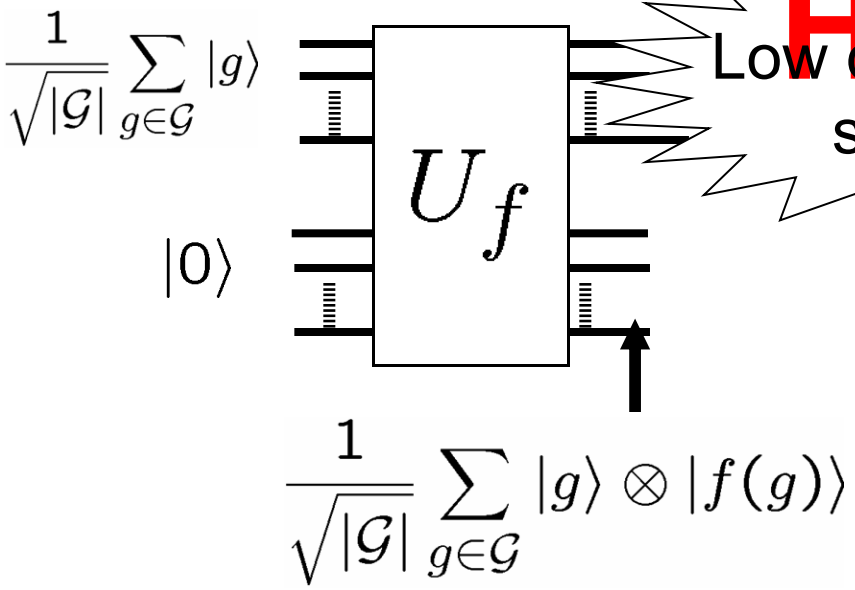
S_4

the Dihedral group → an efficient algorithm for poly(d)–unique shortest vector in a lattice



D8

Dihedral HSP

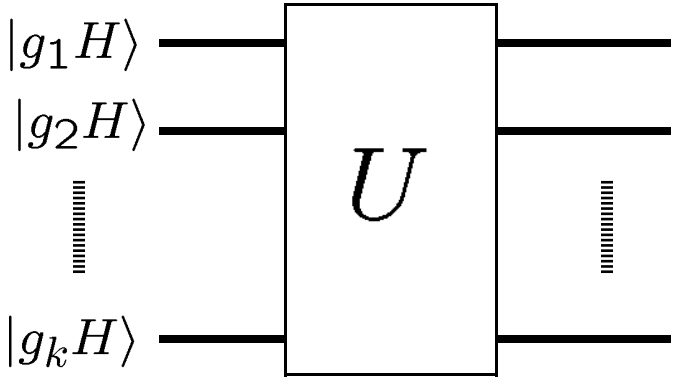


Low dimensional systems

$$|gH\rangle = \frac{1}{\sqrt{|H|}}\sum_{h\in H}|gh\rangle$$

Not much entanglement

k copies of the hidden subgroup states:



entangled measurement to determine the hidden subgroup

Query Complexity of HSP

Theorem: [Ettinger, Hoyer, Knill 99] k=O(log(N)) hidden subgroup states are sufficient to solve the hidden subgroup problem. (but no efficient quantum circuit is known.)

Theorem: [Bacon, Childs, van Dam 05] k=log(N) hidden subgroup states are necessary to solve the dihedral hidden subgroup problem (no known efficient quantum circuit)

1. Measurement across k copies is in entangled basis
2. Success probability of measurement is related

The Subset Sum Problem

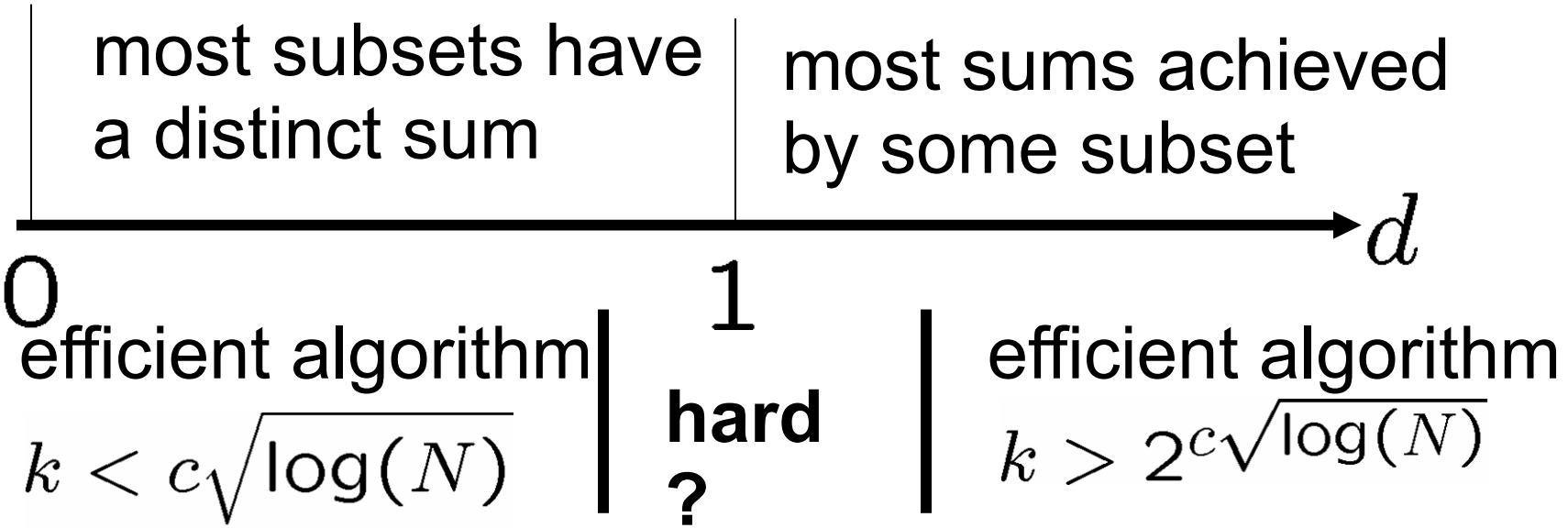
Given: k integers from $\mathbb{Z}_N := \{0, 1, \ldots, N-1\}$

i.e. $x \in \mathbb{Z}_N^k$, and a target $t \in \mathbb{Z}_N$

Problem: Does there exist a subset which sums to t

Define: density $d = \frac{k}{\log(N)}$

"low density" **"high density"**



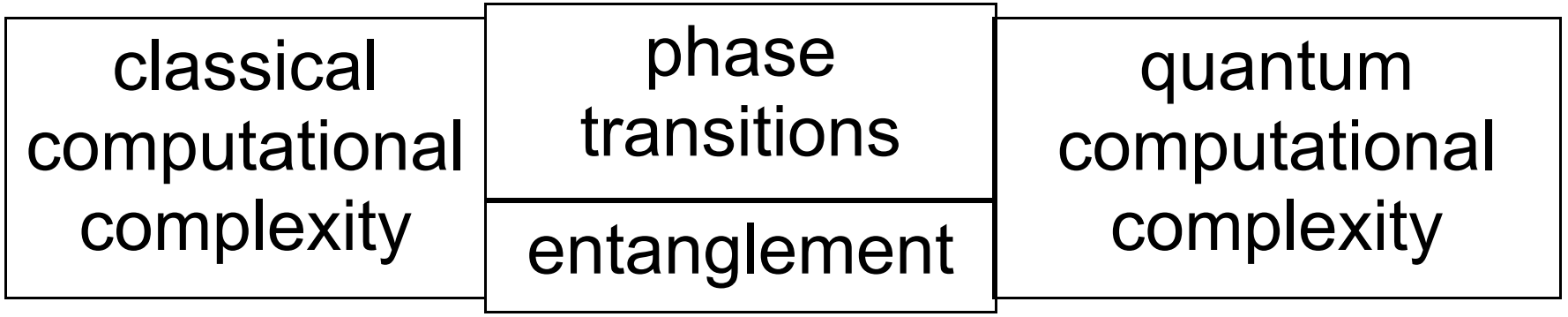
[Lagarias, Odlyzko

[Flaxman, Pryzdatek 04]

Query Complexity of HSP

Theorem: [Bacon, Childs, van Dam 05] k=log(N) hidden subgroup states are necessary to solve the dihedral hidden subgroup problem.

Phase transition in a classical compute science problem appearing in a quantum setting….



Unfortunately, no known efficient circuit….

Towards New HSP Algorithms

We need multiple copies of the hidden subgroup states. $|g_1 H\rangle$

$|g_2 H\rangle$

But we have no examples of efficient circuits which require entangled measurements…

Heisenberg Group

Semidirect product group $\mathbb{Z}_p^2 \rtimes \mathbb{Z}_p$



Elements of the group:

$$\left(\begin{pmatrix} x \\ y \end{pmatrix}, b \right) \quad x, y, b \in \mathbb{Z}_p$$

Multiplication rule:

$$\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, b_1 \right) \left(\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, b_2 \right) =$$

$$\left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}^{b_1} \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, b_1 + b_2 \right)$$

Copenhagen: “Your talent is for ski-ing too fast for anyone to see where you are. For always being in more than one position at a time, like one of your particles.”

Heisenberg HSP

Bacon, Childs, and van Dam FOCS 2005

For one copy, the probability of successfully identifying the hidden subgroup is

$$Pr(succ) = O\left(\frac{1}{p}\right)$$

But for two copies, the probability of successfully identifying the hidden subgroup is

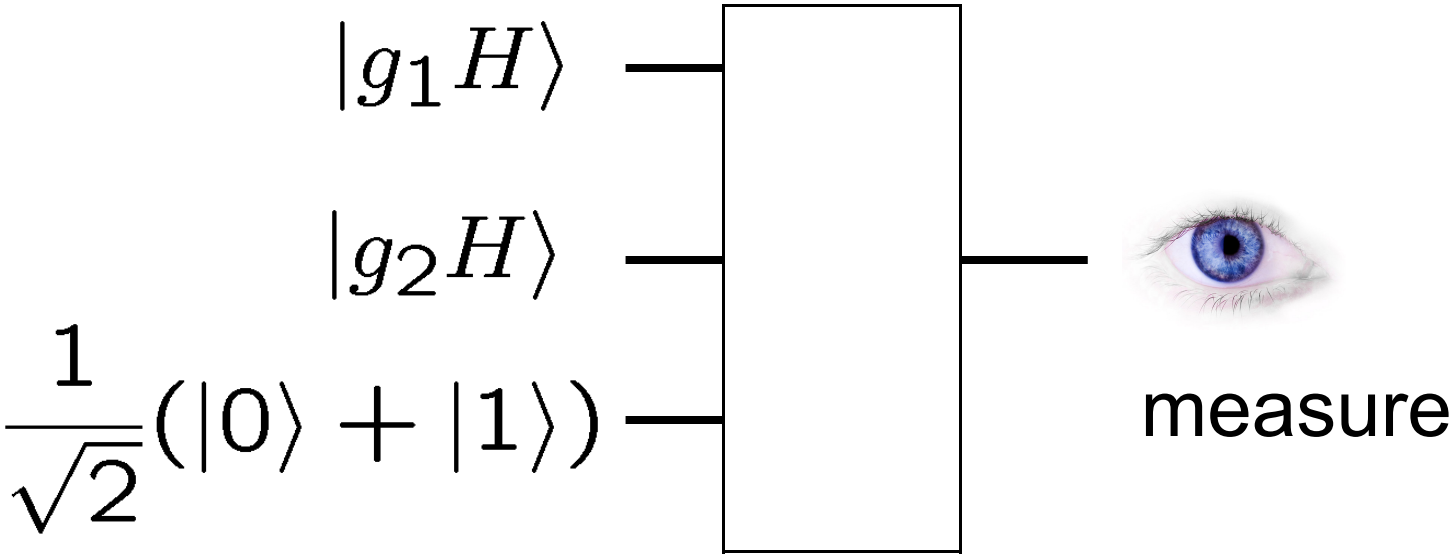
$$Pr(succ) = O(1)$$

This would require an entangled measurement.

Heisenberg HSP Algorithm

Bacon, Childs, and van Dam FOCS 2005

Efficient quantum circuit families



The Hidden Subgroup Success

- Abelian groups [Shor 94; Kitaev 95]
- $\mathbb{Z}_2^n \rtimes \mathbb{Z}_2$ [Püschel, Rötteler, Beth 98]
- Normal subgroups [Hallgren, Russell, Ta-Shma 00]
- "Amost Abelian" groups [GSVV 01]
- $\mathbb{Z}_{p^k}^n \rtimes \mathbb{Z}_2$ [FIMMS 02]
- "Near Hamiltonian" groups [Gavinsky 04]
- Some q-hedral groups [MRRS 04]
- Some [Inui, Le Gall 04]
- Heiser $\mathbb{Z}_p^k \rtimes \mathbb{Z}_p$ e groups [this talk]

$\mathbb{Z}_p^k \rtimes \mathbb{Z}_p$

Bacon, Childs, and van Dam FOCS 2005

Things are heating up

Quantum Algorithms

The power: entanglement is a big part of the story

Leads to new quantum algorithms.

Open Problems

Efficient quantum algorithms for the nonabelian hidden subgroup problem?

We need a deeper understanding of entanglement and and a merging of this understanding with dynamics (interference)

Reasons to Study

1. In the future we may build large quantum computers.

2. The ultimate limits of computation are quantum limits

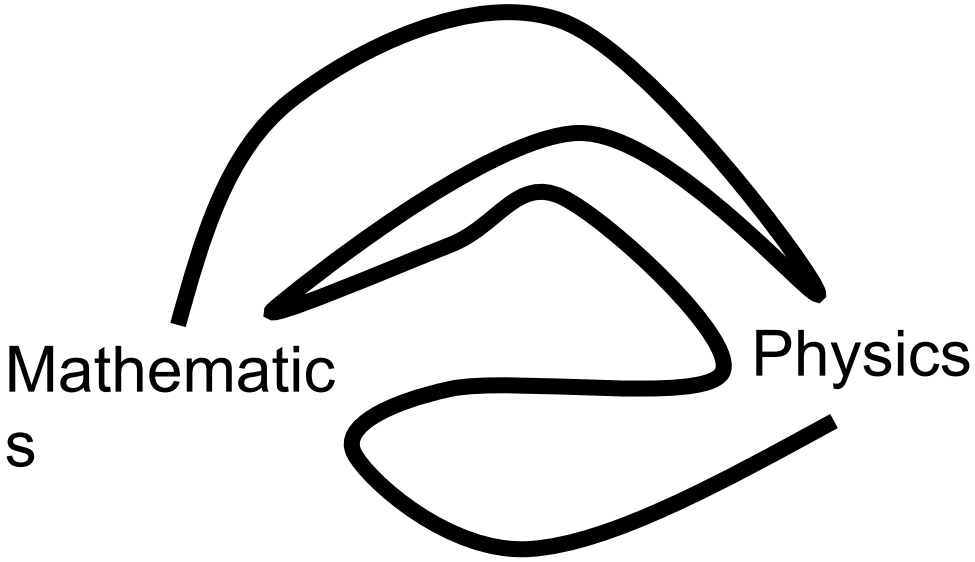
   If someone showed P does not equal NP, would we be satisfied today?

3. The beginning of an interplay between physics and computer science?

Math and Physics

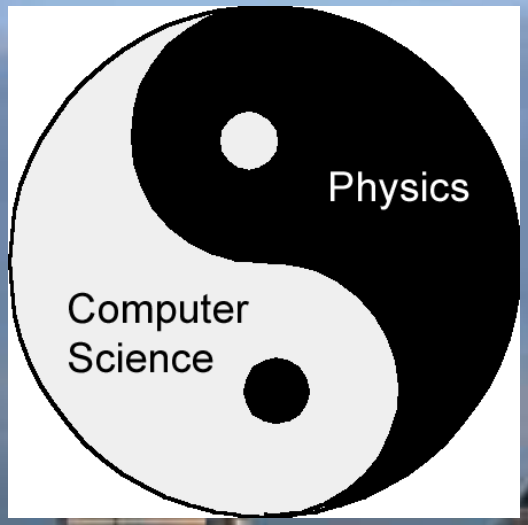
"The must incomprehensible thing about the universe is that it is comprehensible"





Why no similar interplay for the theory of computation?

The Power of Quantum Algorithms





Dave
Ba

Department of Computer Science & Engineering
University of Washington